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(12) **United States Patent**
Takebe et al.(10) **Patent No.:** **US 12,414,967 B2**(45) **Date of Patent:** ***Sep. 16, 2025**(54) **COMPOSITIONS AND METHODS OF
TREATING LIVER DISEASE**USPC 514/169
See application file for complete search history.(71) Applicant: **Children's Hospital Medical Center,**
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Cincinnati, OH (US)(*) Notice: Subject to any disclaimer, the term of this
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U.S.C. 154(b) by 0 days.This patent is subject to a terminal dis-
claimer.(21) Appl. No.: **16/857,338**(22) Filed: **Apr. 24, 2020**(65) **Prior Publication Data**

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on Nov. 4, 2016.(51) **Int. Cl.****A61K 35/407** (2015.01)**A61K 9/00** (2006.01)**A61K 31/575** (2006.01)**A61K 45/06** (2006.01)**A61P 1/16** (2006.01)**C12N 5/071** (2010.01)**C12N 5/074** (2010.01)**C12N 15/01** (2006.01)**G01N 33/50** (2006.01)(52) **U.S. Cl.**CPC **A61K 35/407** (2013.01); **A61K 9/0029**
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US LLP(57) **ABSTRACT**Disclosed are methods of treating or reducing the occurrence
of a steatohepatitis disorder. The disorder may include, for
example, NASH, parenteral nutrition associated liver dis-
ease (PNALD), or genetic forms of liver disease. The
method may comprise the step of administering a compo-
sition comprising obeticholic acid to an individual in need
thereof.

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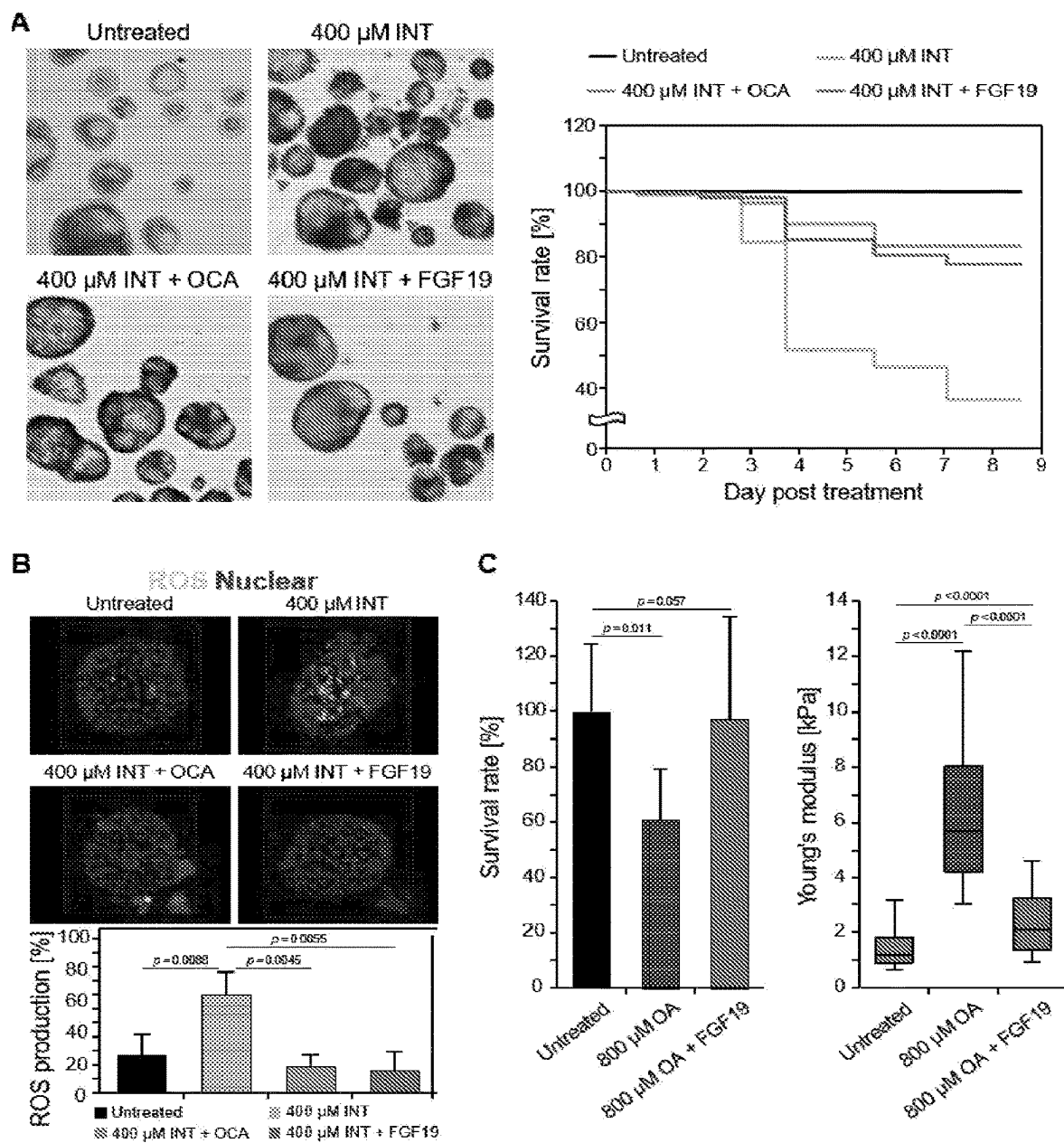


FIG. 1

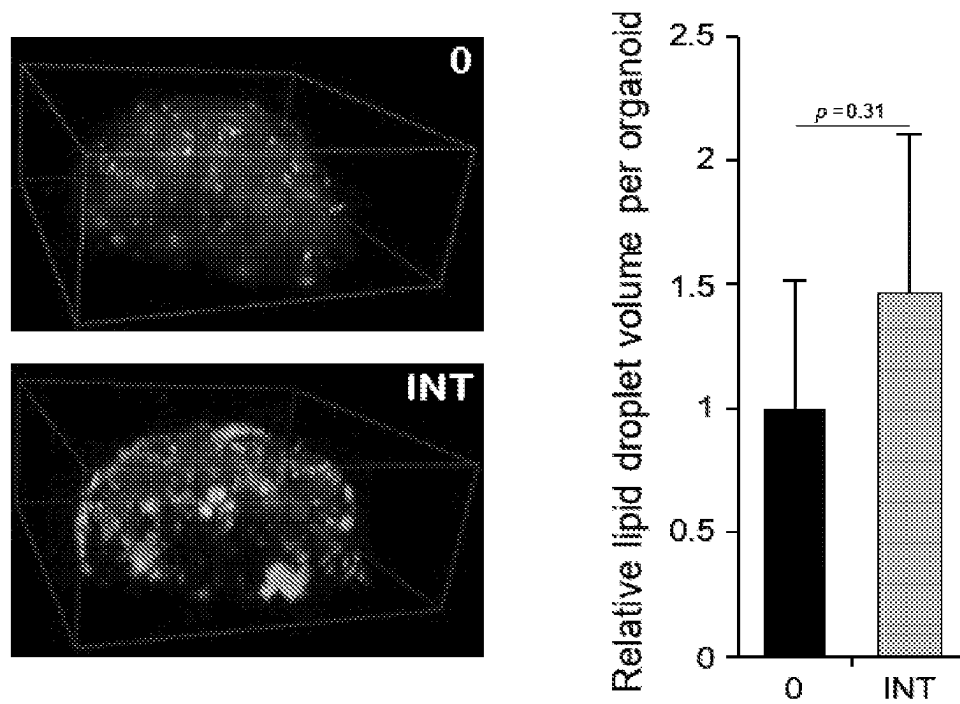


FIG. 2

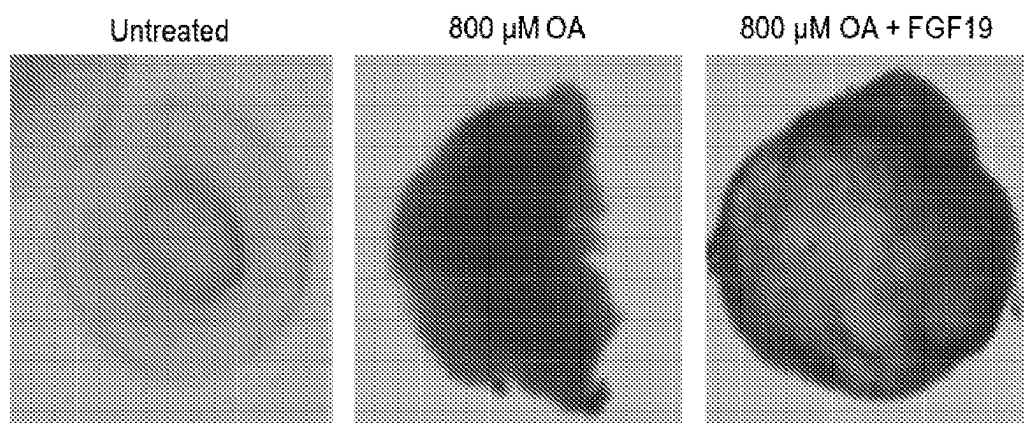


FIG. 3

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COMPOSITIONS AND METHODS OF TREATING LIVER DISEASE

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to and benefit of U.S. Provisional Patent Application No. 62/471,371, filed Nov. 4, 2016, and 62/517,414, filed Jun. 9, 2016, the contents of each are incorporated by reference in their entirety for all purposes.

BACKGROUND

PNALD is the most devastating complication of long-term PN that occurs in the majority of children receiving PN (Kumar and Teckman, 2015; Orso et al., 2016; Zambrano et al., 2004). Because its progression is typically insidious and its long-term consequences are generally underappreciated, PNALD is often recognized too late, when liver injury is irreversible. It is histologically characterized by intrahepatic cholestasis but can progress to fibrosis and cirrhotic liver failure with continued exposure to PN. There are no established ameliorative strategies for PNALD and PNALD is the leading indication for liver transplantation in infants (El Kasmi et al., 2013).

BRIEF SUMMARY

Disclosed are methods of treating or reducing the occurrence of a steatohepatitis disorder. The disorder may include, for example, NASH, parenteral nutrition associated liver disease (PNALD), or genetic forms of liver disease. The method may comprise the step of administering a composition comprising obeticholic acid to an individual in need thereof.

BRIEF DESCRIPTION OF THE DRAWING

Those of skill in the art will understand that the drawings, described below, are for illustrative purposes only. The drawings are not intended to limit the scope of the present teachings in any way.

FIG. 1. Obeticholic acid treatment rescue PNALD like pathology in vitro A. Bright-field image of and survival rate (%) of HLO treated 0 (black line), 400 μ M Intralipid (Int; light blue line), 400 μ M Int with OCA (gray line), and 400 μ M Int with 40 ng/ml FGF19 (pink line). Arrowheads indicate the dead organoids with clumped morphology in the images. Survival rate (%) of 0 (black bar), 800 μ M OA (blue bar), 800 μ M OA+FGF19 (pink bar) at the same time point. The survival of 800 μ M OA HLO was significantly improved by FGF19 addition. B. Live-cell imaging of ROS (green) and nuclear (blue). OCA (gray bar) and FGF19 (pink bar) inhibit the ROS production caused by 400 μ M Int (light blue bar). C. Intralipid. The stiffness of 0 (gray bar), 800 μ M OA (blue bar), 800 μ M OA+FGF19 (pink bar) treated HLO was assessed by AFM. The stiffness of 800 μ M OA HLO was significantly softened by FGF19 addition.

FIG. 2. Lipids accumulation in the organoid in the presence and absence of 400 μ M Intralipid.

FIG. 3. Bright-field image of liver organoids in the culture of non-treatment, 800 μ M of OA alone, and 800 μ M of OA with 40 ng/ml FGF19.

DETAILED DESCRIPTION OF THE INVENTION

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly

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understood by one of ordinary skill in the art. In case of conflict, the present document, including definitions, will control. Preferred methods and materials are described below, although methods and materials similar or equivalent to those described herein can be used in practice or testing of the present invention. All publications, patent applications, patents and other references mentioned herein are incorporated by reference in their entirety. The materials, methods, and examples disclosed herein are illustrative only and not intended to be limiting.

The terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein.

As used herein and in the appended claims, the singular forms “a,” “and,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a method” includes a plurality of such methods and reference to “a dose” includes reference to one or more doses and equivalents thereof known to those skilled in the art, and so forth.

The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the art. Alternatively, “about” can mean a range of up to 20%, or up to 10%, or up to 5%, or up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value. Where particular values are described in the application and claims, unless otherwise stated the term “about” meaning within an acceptable error range for the particular value should be assumed.

The terms “individual,” “host,” “subject,” and “patient” are used interchangeably to refer to an animal that is the object of treatment, observation and/or experiment. Generally, the term refers to a human patient, but the methods and compositions may be equally applicable to non-human subjects such as other mammals. In some embodiments, the terms refer to humans. In further embodiments, the terms refer to children.

“Therapeutically effective amount” relates to the amount or dose of an active compound or composition described herein that will lead to one or more therapeutic effect, in particular, a desired beneficial effect. A therapeutically effective amount of a substance can vary according to factors such as the disease state, age, sex, and weight of the subject, and the ability of the substance to elicit a desired response in the subject. Dosage regime may be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered daily or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation.

The term “parenteral nutrition,” or “intravenous feeding,” is a method of providing nutrition into an individual via intravenous administration. Depending on which vein is used, this procedure is often referred to as either total parenteral nutrition (TPN) or peripheral parenteral nutrition (PPN).

The phrase “pharmaceutically acceptable,” as used in connection with compositions of the disclosure, refers to molecular entities and other ingredients of such composi-

tions that are physiologically tolerable and do not typically produce untoward reactions when administered to a subject (e.g., human) In certain embodiments, as used herein, the term "pharmaceutically acceptable" means approved by a regulatory agency of a Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in mammals (e.g., humans).

The terms "pharmaceutically acceptable salts" or "a pharmaceutically acceptable salt thereof" refer to salts prepared from pharmaceutically acceptable, non-toxic acids or bases. Suitable pharmaceutically acceptable salts include metallic salts, e.g., salts of aluminum, zinc, alkali metal salts such as lithium, sodium, and potassium salts, alkaline earth metal salts such as calcium and magnesium salts; organic salts; salts of free acids and bases; inorganic salts, e.g., sulfate, hydrochloride, and hydrobromide; and other salts which are currently in widespread pharmaceutical use and are listed in sources well known to those of skill in the art, such as The Merck Index. Any suitable constituent can be selected to make a salt of an active drug discussed herein, provided that it is non-toxic and does not substantially interfere with the desired activity. In addition to salts, pharmaceutically acceptable precursors and derivatives of the compounds can be employed. Pharmaceutically acceptable amides, lower alkyl esters, and protected derivatives of the disclosed actives can also be suitable for use in the compositions and methods disclosed herein. A salt of a compound of this disclosure may be formed between an acid and a basic group of the compound, such as an amino functional group, or a base and an acidic group of the compound, such as a carboxyl functional group. According to another embodiment, the compound is a pharmaceutically acceptable acid addition salt. Acids commonly employed to form pharmaceutically acceptable salts include inorganic acids such as hydrogen bisulfide, hydrochloric acid, hydrobromic acid, hydroiodic acid, sulfuric acid and phosphoric acid, as well as organic acids such as para-toluenesulfonic acid, salicylic acid, tartaric acid, bitartaric acid, ascorbic acid, maleic acid, besylic acid, fumaric acid, gluconic acid, glucuronic acid, formic acid, glutamic acid, methanesulfonic acid, ethanesulfonic acid, benzenesulfonic acid, lactic acid, oxalic acid, para-bromophenylsulfonic acid, carbonic acid, succinic acid, citric acid, benzoic acid and acetic acid, as well as related inorganic and organic acids. Such pharmaceutically acceptable salts thus include sulfate, pyrosulfate, bisulfate, sulfite, bisulfite, phosphate, monohydrogenphosphate, dihydrogenphosphate, metaphosphate, pyrophosphate, chloride, bromide, iodide, acetate, propionate, decanoate, caprylate, acrylate, formate, isobutyrate, caprate, heptanoate, propionate, oxalate, malonate, succinate, suberate, sebacate, fumarate, maleate, butyne-1,4-dioate, hexyne-1,6-dioate, benzoate, chlorobenzoate, methylbenzoate, dinitrobenzoate, hydroxybenzoate, methoxybenzoate, phthalate, terephthalate, sulfonate, xylene sulfonate, phenylacetate, phenylpropionate, phenylbutyrate, citrate, lactate, β -hydroxybutyrate, glycolate, maleate, tartrate, methanesulfonate, propane-sulfonate, naphthalene-1-sulfonate, naphthalene-2-sulfonate, mandelate and other salts. In one embodiment, pharmaceutically acceptable acid addition salts include those formed with mineral acids such as hydrochloric acid and hydrobromic acid, and especially those formed with organic acids such as maleic acid.

The terms "treat," "treating" or "treatment," as used herein, refers to methods of alleviating, abating or ameliorating a disease or condition symptoms, preventing additional symptoms, ameliorating or preventing the underlying metabolic causes of symptoms, inhibiting the disease or

condition, arresting the development of the disease or condition, relieving the disease or condition, causing regression of the disease or condition, relieving a condition caused by the disease or condition, or stopping the symptoms of the disease or condition either prophylactically and/or therapeutically.

The term "carrier" applied to pharmaceutical compositions of the disclosure refers to a diluent, excipient, or vehicle with which an active compound is administered. Such pharmaceutical carriers can be sterile liquids, such as water, saline solutions, aqueous dextrose solutions, aqueous glycerol solutions, and oils, including those of petroleum, animal, vegetable, or synthetic origin, such as peanut oil, soybean oil, mineral oil, sesame oil and the like. Suitable pharmaceutical carriers are described in "Remington's Pharmaceutical Sciences" (any edition).

The term "compound," as used herein, is also intended to include any salts, solvates, or hydrates thereof.

Applicant has discovered novel methods for the treatment of liver disease in an individual in need thereof, which may include administration of a disclosed compound to an individual in need thereof.

In one aspect, a method of treating or reducing the occurrence of a steatohepatitis disorder is disclosed. The disorder may include, for example, NASH, parenteral nutrition associated liver disease (PNALD), or genetic forms of liver disease. The method may comprise the step of administering a composition comprising obeticholic acid to an individual in need thereof.

In one aspect, the composition may be administered in an amount sufficient to reduce or prevent the occurrence of liver cell fibrosis.

In one aspect, the composition may be administered prior to, following, or during administration of parenteral nutrition to said individual in need thereof.

In one aspect, a composition is disclosed, wherein the composition may comprise total parenteral nutrition (TPN) composition or peripheral parenteral nutrition (PPN) composition; and obeticholic acid.

In a further aspect, an article of manufacture is disclosed. The article of manufacture may comprise a container, a composition as described herein, the composition being in a dosage form, and instructions for administering the dosage form to an individual diagnosed or suspected of having or developing a liver disease as disclosed herein.

Dosage

In one aspect, an agent disclosed herein may be present in an amount of from about 0.5% to about 95%, or from about 1% to about 90%, or from about 2% to about 85%, or from about 3% to about 80%, or from about 4%, about 75%, or from about 5% to about 70%, or from about 6%, about 65%, or from about 7% to about 60%, or from about 8% to about 55%, or from about 9% to about 50%, or from about 10% to about 40%, by weight of the composition.

The compositions may be administered in oral dosage forms such as tablets, capsules (each of which includes sustained release or timed release formulations), pills, powders, granules, elixirs, tinctures, suspensions, syrups, and emulsions. They may also be administered in intravenous (bolus or infusion), intraperitoneal, subcutaneous, intral-lesional, or intramuscular forms all utilizing dosage forms well known to those of ordinary skill in the pharmaceutical arts. The compositions may be administered by intranasal route via topical use of suitable intranasal vehicles, or via a transdermal route, for example using conventional transdermal skin patches. A dosage protocol for administration using

a transdermal delivery system may be continuous rather than intermittent throughout the dosage regimen.

In one aspect, the compounds may be administered at the rate of 100 µg to 1000 mg per day per kg of body weight. Orally, the compounds may be suitably administered at the rate of about 100, 150, 200, 250, 300, 350, 400, 450, or 500 µg to about 1, 5, 10, 25, 50, 75, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000 mg per day per kg of body weight. The required dose can be administered in one or more portions. For oral administration, suitable forms are, for example, tablets, gel, aerosols, pills, dragees, syrups, suspensions, emulsions, solutions, powders and granules; one method of administration includes using a suitable form containing from 1 mg to about 500 mg of active substance. In one aspect, administration may comprise using a suitable form containing from about 1, 2, 5, 10, 25, or 50 mg to about 100, 200, 300, 400, 500 mg of active substance.

A dosage regimen will vary depending upon known factors such as the pharmacodynamic characteristics of the agents and their mode and route of administration; the species, age, sex, health, medical condition, and weight of the patient, the nature and extent of the symptoms, the kind of concurrent treatment, the frequency of treatment, the route of administration, the renal and hepatic function of the patient, and the desired effect. The effective amount of a drug required to prevent, counter, or arrest progression of a symptom or effect of a muscle contracture can be readily determined by an ordinarily skilled physician.

The pharmaceutical compositions may include suitable dosage forms for oral, parenteral (including subcutaneous, intramuscular, intradermal and intravenous), transdermal, sublingual, bronchial or nasal administration. Thus, if a solid carrier is used, the preparation may be tableted, placed in a hard gelatin capsule in powder or pellet form, or in the form of a troche or lozenge. The solid carrier may contain conventional excipients such as binding agents, fillers, tableting lubricants, disintegrants, wetting agents and the like. The tablet may, if desired, be film coated by conventional techniques. Oral preparations include push-fit capsules made of gelatin, as well as soft, sealed capsules made of gelatin and a coating, such as glycerol or sorbitol. Push-fit capsules can contain active ingredients mixed with a filler or binders, such as lactose or starches, lubricants, such as talc or magnesium stearate, and, optionally, stabilizers. In soft capsules, the active compounds may be dissolved or suspended in suitable liquids, such as fatty oils, liquid, or liquid polyethylene glycol with or without stabilizers. If a liquid carrier is employed, the preparation may be in the form of a syrup, emulsion, soft gelatin capsule, sterile vehicle for injection, an aqueous or non-aqueous liquid suspension, or may be a dry product for reconstitution with water or other suitable vehicle before use. Liquid preparations may contain conventional additives such as suspending agents, emulsifying agents, wetting agents, non-aqueous vehicle (including edible oils), preservatives, as well as flavoring and/or coloring agents. For parenteral administration, a vehicle normally will comprise sterile water, at least in large part, although saline solutions, glucose solutions and like may be utilized. Injectable suspensions also may be used, in which case conventional suspending agents may be employed. Conventional preservatives, buffering agents and the like also may be added to the parenteral dosage forms. For topical or nasal administration, penetrants or permeation agents that are appropriate to the particular barrier to be permeated are used in the formulation. Such penetrants are generally known in the art. The pharmaceutical compositions are prepared by conventional techniques appropriate to

the desired preparation containing appropriate amounts of the active ingredient, that is, one or more of the disclosed active agents or a pharmaceutically acceptable salt thereof according to the invention.

The dosage of an agent disclosed herein used to achieve a therapeutic effect will depend not only on such factors as the age, weight and sex of the patient and mode of administration, but also on the degree of inhibition desired and the potency of an agent disclosed herein for the particular disorder or disease concerned. It is also contemplated that the treatment and dosage of an agent disclosed herein may be administered in unit dosage form and that the unit dosage form would be adjusted accordingly by one skilled in the art to reflect the relative level of activity. The decision as to the particular dosage to be employed (and the number of times to be administered per day) is within the discretion of the physician, and may be varied by titration of the dosage to the particular circumstances of this invention to produce the desired therapeutic effect.

Routes of Administration

Any suitable route of administration can be employed for providing the patient with an effective dosage of the disclosed compositions. For example, oral, rectal, transdermal, parenteral (subcutaneous, intramuscular, intravenous), intrathecal, topical, inhalable, and like forms of administration can be employed. Suitable dosage forms include tablets, troches, dispersions, suspensions, solutions, capsules, patches, and the like. Administration of medicaments prepared from the compounds described herein can be by any suitable method capable of introducing the compounds into the bloodstream. In some embodiments, the formulations can contain a mixture of active compounds with pharmaceutically acceptable carriers or diluents known to those of skill in the art.

The compositions can be prepared in any desired form, for example, tables, powders, capsules, injectables, suspensions, sachets, cachets, patches, solutions, elixirs, and aerosols. Carriers such as starches, sugars, microcrystalline cellulose, diluents, granulating agents, lubricants, binders, disintegrating agents, and the like can be used in oral solid preparations. In certain embodiments, the compositions are prepared as oral solid preparations (such as powders, capsules, and tablets). In certain embodiments, the compositions are prepared as oral liquid preparations. In some embodiments, the oral solid preparations are tablets. If desired, tablets can be coated by standard aqueous or non-aqueous techniques.

In addition to the dosage forms set out above, one or more disclosed compounds may also be administered by sustained release, delayed release, or controlled release compositions and/or delivery devices.

Pharmaceutical compositions suitable for oral administration can be provided as discrete units such as capsules, cachets, sachets, patches, injectables, tablets, and aerosol sprays, each containing predetermined amounts of the active ingredients, as powder or granules, or as a solution or a suspension in an aqueous liquid, a non-aqueous liquid, an oil-in-water emulsion, or a water-in-oil liquid emulsion. Such compositions can be prepared by any of the conventional methods of pharmacy, and may include the step of bringing into association the active ingredients with a carrier which constitutes one or more ingredients. In general, the compositions may be prepared by uniformly and intimately admixing the active ingredients with liquid carriers, finely divided solid carriers, or both, and then, optionally, shaping the product into the desired presentation.

For example, a tablet can be prepared by compression or molding, optionally, with one or more additional ingredients. Compressed tablets can be prepared by compressing in a suitable machine the active ingredient in a free-flowing form such as powder or granules, optionally mixed with a binder, lubricant, inert diluent, surface active or dispersing agent. Molded tablets can be made by molding, in a suitable machine, a mixture of the powdered compound moistened with an inert liquid diluent.

A composition or formulation may be administered to a subject continuously or periodically.

The compositions or fractions thereof typically comprise suitable pharmaceutical diluents, excipients, vehicles, or carriers selected based on the intended form of administration, and consistent with conventional pharmaceutical practices. The carriers, vehicles etc. may be adapted to provide an additive, synergistically effective or therapeutically effective amount of the active compounds. Suitable pharmaceutical diluents, excipients, vehicles, and carriers are described in the standard text, Remington: The Science and Practice of Pharmacy, and in The United States Pharmacopeia: The National Formulary (USP 24 NF 19) published in 1999. By way of example, for oral administration in the form of a capsule or tablet, the active components can be combined with an oral, non-toxic pharmaceutically acceptable inert carrier such as lactose, starch, sucrose, methyl cellulose, magnesium stearate, glucose, calcium, sulfate, dicalcium phosphate, mannitol, sorbitol, and the like. For oral administration in a liquid form, the agents may be combined with any oral, non-toxic, pharmaceutically acceptable inert carrier such as ethanol, glycerol, water, and the like. Suitable binders (e.g. gelatin, starch, corn sweeteners, natural sugars including glucose; natural and synthetic gums, and waxes), lubricants (e.g. sodium oleate, sodium stearate, magnesium stearate, sodium benzoate, sodium acetate, and sodium chloride), disintegrating agents (e.g. starch, methyl cellulose, agar, bentonite, and xanthan gum), flavoring agents, and coloring agents may also be combined in the compositions or components thereof.

In one aspect, a pharmaceutical composition may have pH from about 7 to 10.

Formulations for parenteral administration of a composition may include aqueous solutions, syrups, aqueous or oil suspensions and emulsions with edible oil such as cottonseed oil, coconut oil or peanut oil. Dispersing or suspending agents that can be used for aqueous suspensions include synthetic or natural gums, such as tragacanth, alginate, acacia, dextran, sodium carboxymethylcellulose, gelatin, methylcellulose, and polyvinylpyrrolidone.

Compositions for parenteral administration may include sterile aqueous or non-aqueous solvents, such as water, isotonic saline, isotonic glucose solution, buffer solution, or other solvents conveniently used for parenteral administration of therapeutically active agents. A composition intended for parenteral administration may also include conventional additives such as stabilizers, buffers, or preservatives.

In an embodiment, a solid form pharmaceutical composition is provided (e.g. tablets, capsules, powdered, or pulverized form) comprising one or more disclosed compounds or salt thereof.

After pharmaceutical compositions have been prepared, they can be placed in an appropriate container and labeled for treatment of an indicated condition. For administration of a composition, such labeling would include amount, frequency, and method of administration.

Kits

Kits are also provided. In one aspect, a kit may comprise or consist essentially of agents or compositions described herein. The kit may be a package that houses a container which may contain a composition comprising an oxime or pharmaceutically acceptable salt thereof as disclosed herein, and also houses instructions for administering the agent or composition to a subject. In one aspect, a pharmaceutical pack or kit is provided comprising one or more containers filled with one or more composition as disclosed herein. Associated with such container(s) can be various written materials such as instructions for use, or a notice in the form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals or biological products, which notice reflects approval by the agency of manufacture, use, or sale for human administration.

As there may be advantages to mixing a component of a composition described herein and a pharmaceutically acceptable carrier, excipient or vehicle near the time of use, kits in which components of the compositions are packaged separately are disclosed. For example, the kit can contain an active ingredient in a powdered or other dry form in, for example, a sterile vial or ampule and, in a separate container within the kit, a carrier, excipient, or vehicle, or a component of a carrier, excipient, or vehicle (in liquid or dry form). In one aspect, the kit can contain a component in a dry form, typically as a powder, often in a lyophilized form in, for example, a sterile vial or ampule and, in a separate container within the kit, a carrier, excipient, or vehicle, or a component of a carrier, excipient, or vehicle. Alternatively, the kit may contain a component in the form of a concentrated solution that is diluted prior to administration. Any of the components described herein, any of the carriers, excipients or vehicles described herein, and any combination of components and carriers, excipients or vehicles can be included in a kit.

Optionally, a kit may also contain instructions for preparation or use (e.g., written instructions printed on the outer container or on a leaflet placed therein) and one or more devices to aid the preparation of the solution and/or its administration to a patient (e.g., one or a plurality of syringes, needles, filters, tape, tubing, alcohol swabs, or the like). Compositions which are more concentrated than those administered to a subject can be prepared. Accordingly, such compositions can be included in the kits with, optionally, suitable materials (e.g., water, saline, or other physiologically acceptable solutions) for dilution. Instructions included with the kit can include, where appropriate, instructions for dilution.

In other embodiments, the kits can include pre-mixed compositions and instructions for solubilizing any precipitate that may have formed during shipping or storage. Kits containing solutions of one or more of the aforementioned active agents, or pharmaceutically acceptable salts thereof, and one or more carriers, excipients or vehicles may also contain any of the materials mentioned above (e.g., any device to aid in preparing the composition for administration or in the administration per se). The instructions in these kits may describe suitable indications (e.g., a description of patients amenable to treatment) and instructions for administering the solution to a patient.

EXAMPLES

Repurposing OCA for PNALD Therapy

In this study, Applicant found that OCA and FGF19 inhibited ROS production caused by Intralipid exposure in HLO, leading to an improvement in survival rate of HLO.

Moreover, the stiffness of OA treated HLO, measured by AFM, was decreased by FGF19 exposure. These results provide direct experimental evidence that OCA and FGF19 play a preventive role in the pathogenesis of PNALD organoid model and that the suppression of ROS production and fibrosis by FXR agonism is the likely mechanism of protection against PNALD. Our study thus highlighted the potential repurposing of OCA for the treatment of patients with PNALD.

Obeticholic Acid Treatment Improved PNALD-Like Pathology of HLO

PNALD is an iatrogenic fatty liver disease which occurs frequently in infants provided with parental nutrition (PN) solutions employing a lipid emulsion (Intralipid) that provide life-sustaining calories in the setting of inadequate absorption of enteral nutrients (Kumar and Teckman, 2015; Orso et al., 2016). PNALD has a histological evidence of steatosis, cell death (apoptosis), and fibrosis (Nandivada et al., 2013). To recapitulate PNALD pathology in HLO, we reconstituted components of Intralipid and placed it directly into HLO culture. Similar to OA treatment, significant lipid accumulation was confirmed in 400 μ M Intralipid treated HLO reflecting a steatosis-like condition (FIG. 2). In parallel, we established an organoid survival assessment algorithm by morphological analysis backed by viability assessment (FIGS. 3 and 2, panel A.) and demonstrated that HLO survival of 400 μ M Intralipid treated group was severely compromised by 37% after 12 days of culture (FIG. 2, panel A).

Because there are no established ameliorative strategies, and PNALD is the leading indication for liver transplantation in infants (El Kasmi et al., 2013), we further investigated preventive and therapeutic effects of FXR agonist obeticholic acid (OCA) and FGF19 hormone that are activated by OCA, which have shown great potential in nonalcoholic steatohepatitis treatment (Dash et al., 2017; Kumar and Teckman, 2015; Neuschwander-Tetri et al., 2015). To implicate the therapeutic significance of OCA and FGF19 for PNALD like pathology in HLO, OCA and FGF19 were added to 400 μ M Intralipid exposed HLO. By addition of OCA and FGF19 into PNALD-HLO, the number of surviving organoids remarkably improved around 83 and 78% in total, respectively. Mechanistically, the accumulation of reactive oxygen species (ROS) is strongly associated with the fatty liver related hepatocyte death (Nandivada et al., 2013; Tsedensodnom and Sadler, 2013). We thus wondered whether OCA and FGF19 would inhibit the production of ROS so we sought to confirm that theory with live-cell imaging with CellROX dye. Consistent with HLO survival improvements, ROS production was 60% in 400 μ M Intralipid treated HLO but reduced to 20 and 30% by OCA and FGF19 treatments, respectively (FIG. 7, B), suggesting that OCA and FGF19 might have therapeutic potential against PNALD through suppressing ROS production.

To further determine whether OCA and FGF19 have an impact on fibrosis, HLO stiffness was assessed. Organoid damage induced by Intralipid is extremely significant and treated HLOs for 5 days are not able to be stabilized on a dish, which is a minimal requirement for assessing the rigidity by AFM. To circumvent this limitation, the stiffness was assessed at Dya3. These observations indicate that OCA and FGF19 reduce the progression of fibrosis, potentially alleviating PNALD phenotype in an organoid.

Methods

hiPSCs Maintenance. The TkDA3 human iPSC clone used in this study was kindly provided by K. Eto and H. Nakauchi. Human iPSC lines were maintained as described

previously (Takebe et al., 2015; Takebe et al., 2014). Undifferentiated hiPSCs were maintained on feeder-free conditions in mTeSR1 medium (StemCell technologies, Vancouver, Canada) on plates coated with Matrigel (Corning Inc., NY, USA) at 1/30 dilution at 37° C. in 5% CO₂ with 95% air.

Definitive endoderm induction. Human iPSCs into definitive endoderm was differentiated using previously described methods with slight modifications (Spence et al., 2011). In brief, colonies of human iPSCs were isolated in Accutase (Thermo Fisher Scientific Inc., MA, USA) and 150,000 cells/mL, were plated on Matrigel coated tissue culture plate (VWR Scientific Products, West Chester, PA). Medium was changed to RPMI 1640 medium (Life Technologies) containing 100 ng/mL Activin A (R&D Systems, MN, USA) and 50 ng/mL bone morphogenetic protein 4 (BMP4; R&D Systems) at day 1, 100 ng/mL Activin A and 0.2% fetal calf serum (FCS; Thermo Fisher Scientific Inc.) at day 2 and 100 ng/mL Activin A and 2% FCS at day 3. Day 4-6 cells were cultured in Advanced DMEM/F12 (Thermo Fisher Scientific Inc.) with B27 (Life Technologies) and N2 (Gibco, CA, USA) containing 500 ng/ml fibroblast growth factor (FGF4; R&D Systems) and 3 μ M CHIR99021 (Stemgent, MA, USA). Cells were maintained at 37° C. in 5% CO₂ with 95% air and the medium was replaced every day. Spheroids appeared on the plate at day 7 of differentiation.

HLO induction. At day 7, spheroids and attached cells are gently pipetted to be delaminated from dishes. They were centrifuged at 800 rpm for 3 minutes, embedded in a 100% Matrigel drop on the dishes in Advanced DMEM/F12 with B27, N2 and 2 μ M retinoic acid (RA; Sigma, MO, USA) after removing supernatant, and cultured for 4 days. After RA treatment, spheroids embedded in the Matrigel drop were cultured in Hepatocytes Culture Medium (HCM; Lonza, MD, USA) with 10 ng/mL hepatocyte growth factor (HGF; PeproTech, NJ, USA), 0.1 μ M Dexamethasone (Dex; Sigma) and 20 ng/mL Oncostatin M (OSM; R&D Systems). Cultures for HLO induction were maintained at 37° C. in 5% CO₂ with 95% air and the medium was replaced every 2-3 days. To analyze HLO (day 20-30), organoids were isolated from Matrigel by scratching and pipetting.

Albumin, IL-6, and P3NP ELISA. To measure albumin secretion level of HLO, 200 μ L of culture supernatant was collected from HLO embedded in Matrigel. For IL-6 and P3NP, 20-30 organoids were seeded and cultured on an ultra-low attachment multiwell plates 96 well plate (Corning). To define the exact number of organoids in each well and lastly normalize the secreted level for IL-6 and P3NP by the number, the organoids were captured on The KEYENCE BZ-X710 Fluorescence Microscope. The culture supernatants were collected at 24 hrs (for albumin), 96 hrs (for IL-6) and 120 hrs (P3NP) time points after the culture and stored at -80° C. until use. The supernatant was centrifuged at 1,500 rpm for 3 min and to pellet debris, and the resulting supernatant was assayed with Human Albumin ELISA Quantitation Set (Bethyl Laboratories, Inc., TX, USA), Human IL-6 ELISA Kit (Biolegend, CA, USA), and Human N-terminal procollagen III propeptide, PIIINP ELISA Kit (My BioSource, CA, USA) according to the manufacturer's instructions. Significance testing was conducted by Student's t-test.

Bile transport activity. Fluorescein diacetate was used for evaluating bile transport activity in organoids. 10 mg/mL fluorescein diacetate (Sigma) was added into HCM media cultured with HLO and allowed to sit for 5 minutes and captured using fluorescent microscopy BZ-X710 (Keyence, Osaka, Japan).

Phagocyte, lipids, ROS live-cell imaging. After being cultured in an ultra-low attachment 6 multi-well plate, 5-10 HLO were picked up and seeded in a Microslide 8 Well Glass Bottom plate (Ibidi, WI, USA) and subjected to live-cell staining. The following antibodies were used: pHrodo® Red *S. aureus* Bioparticles® Conjugate for Phagocyte activity (Thermo Fisher Scientific Inc.), BODIPY® 493/503 for lipids (Thermo Fisher Scientific Inc.), Di-8-ANEPPS for membrane (Thermo Fisher Scientific Inc.), and CellROX green reagent for ROS detection (Thermo Fisher Scientific Inc.). Nuclear staining was marked by NucBlue Live ReadyProbes Reagent (Thermo Fisher Scientific Inc.). HLO was visualized and scanned on a Nikon A1 Inverted Confocal Microscope (Japan) using 60× water immersion objectives. The final lipid droplet volume was calculated by IMARIS8 and normalized by each organoid size. Significance testing for lipid droplet volume and ROS production (%) was conducted by Student's t-test.

HE staining and immunohistochemistry. HLO were isolated from Matrigel and fixed in 4% paraformaldehyde and embedded in paraffin. Sections were subjected to HE and immunohistochemical staining. The following primary antibodies were used: anti-alpha smooth muscle actin antibody (1:200 dilution; abcam, Cambridge, UK), Desmin antibody (Pre-diluted; Roche, Basel, Switzerland), and CD68 antibody (1:200 dilution; abcam).

Flow cytometry. HLO were isolated from 10 Matrigel droplets and washed by 1×PBS. HLO were dissociated to single cells by the treatment of Trypsin-EDTA (0.05%), phenol red (Gibco) for 10 min. After PBS wash, the single cells were subjected to flow cytometry with BV421-conjugated Epcam antibody (BioLegend), PE-conjugated CD166 antibody (eBioscience, CA, USA), and PE/Cy7-CD68 (eBioscience). DNA was measured by propidium iodide staining.

LPS and FFA exposure and OCA and FGF19 treatment. 20-30 hLO/HLO, which had been isolated from Matrigel and washed by 1×PBS, were divided into each condition and cultured on an ultra-low attachment 6 multi-well plates (Corning). HLO were cultured with LPS (Sigma), OA (Sigma), LA (Sigma), SA (Sigma), or PA (Sigma) and collected at day 1 and 3 (for LPS HLO) and at day 3 and 5 (for OA) after the culture. To test the inhibitory effect of OCA (INT-747, MedChem Express, NJ, USA) and human FGF19 recombinant (Sigma) on HLO, 20-30 HLO were cultured in HCM media in the presence or absence of oleic acid (800 μM), and 1 μM OCA and 40 ng/ml FGF19 were added into 800 μM OA condition. HLO were collected at day 3 for lipids live-cell imaging and at day 5 for stiffness measurement.

Whole mount immunofluorescence. HLO were fixed for 30 min in 4% paraformaldehyde and permeabilized for 15 min with 0.5% Nonidet P-40. HLO were washed by 1×PBS three times and incubated with blocking buffer for 1 h at room temperature. HLO were then incubated with primary antibody; anti-alpha smooth muscle actin antibody (1:50 dilution; abcam) overnight at 4° C. HLO were washed by 1×PBS and incubated in secondary antibody in blocking buffer for 30 min at room temperature. HLO were washed and mounted using Fluoroshield mounting medium with DAPI (abcam). The stained HLO were visualized and scanned on a Nikon A1 Inverted Confocal Microscope (Japan) using 60× water immersion objectives.

RNA isolation, RT-qPCR. RNA was isolated using the RNeasy mini kit (Qiagen, Hilden, Germany). Reverse transcription was carried out using the SuperScriptIII First-Strand Synthesis System for RT-PCR (Invitrogen, CA,

USA) according to manufacturer's protocol. qPCR was carried out using TaqMan gene expression master mix (Applied Biosystems) on a QuantStudio 3 Real-Time PCR System (Thermo Fisher Scientific Inc.). All primers and probe information for each target gene was obtained from the Universal ProbeLibrary Assay Design Center (<https://qpcr.probefinder.com/organism.jsp>). Significance testing was conducted by Student's t-test.

HLO stiffness measurement by AFM. HLOs treated with 0, 50200, 1400, 2800 ng/mL μM LPSOA were used for stiffness measurement with a AFM (NanoWizard IV, JPK Instruments, Germany). The AFM head with a silicon nitride cantilever (CSC37, k=0.3 N/m, MikroMasch, Bulgaria) was mounted on a fluorescence stereo microscope (M205 FA, Leica, Germany) coupled with a Z-axis piezo stage (JPK CellHesion module, JPK Instruments, Germany), which allows the indentation measurement up to the depth of ~100 μm. As a substrate for organoids, a fibronectin-coated dish was used. The tissue culture dish (φ=34 mm, TPP Techno Plastic Products, Switzerland) was first incubated with a 1 μg/mL fibronectin solution (Sigma) at 4° C. for overnight. Then, the tissue culture dish was washed twice by distilled water and dried for 1 hour. Thereafter, HLOs incubated with OALPS for 51 days were deposited to the fibronectin-coated dish and incubated for 1 hour at 37° C. The sample dish was then placed onto the AFM stage, and force-distance curves in a 14×14 matrix in a 25×25 μm square were measured from each HLO. Finally, Young's moduli (E, Pa) of HLOs were determined by fitting the obtained force-distance curves with the modified Hertz model (Sneddon, 1965). Dunn-Holland-Wolfe test was performed for significance testings.

THP-1 cell migration assay. THP-1 cell, which was gifted from T. Suzuki, was maintained in Advanced DMEM/F12 (Thermo Fisher Scientific Inc.) containing 10% FBS. THP-1 floating cells were collected, and 200,000 cells were added with serum-free Advanced DMEM/F12 to the membrane chamber of the CytoSelect™96-Well Cell Migration Assay (5 μm, Fluorometric Format; Cell Biolabs, CA, USA). 10-20 HLO were cultured in HCM media including 0, 400, 800 μM OA with an ultra-low attachment 96 multi-well plate (Corning) for three days. To define the exact number of organoids in each well and lastly normalize the final migrated cells by the number, the organoids were captured on The KEYENCE BZ-X710 Fluorescence Microscope. 150 μL of culture supernatant of HLO was collected and added to the feeder tray of the kit. The kit was incubated at 37° C. for 24 h in a 5% CO₂ cell culture incubator. Cells that had migrated were counted using Countess II FL Automated Cell Counter (Thermo Fisher Scientific Inc.). Significance testing was conducted by Student's t-test.

Triglyceride assay. For quantitative determination of triglycerides, HLO were isolated from one Matrigel drop and divided into HCM media in the presence or absence of oleic acid (800 μM). They were cultured on an ultra-Low attachment 6 multi-well plate for three days. Quantitative estimation of hepatic triglyceride accumulation was performed by an enzymatic assay of triglyceride mass using the Enzy-Chrom Triglyceride assay kit (Bioassay Systems, CA, USA).

HLO survival assay. HLO were collected from Matrigel and washed by 1×PBS. 30-40 organoids were cultured on an ultra-low attachment 6 multi-well plate (Corning). HLO were captured on The KEYENCE BZ-X710 Fluorescence Microscope every day. The surviving and dead organoids were counted manually from the photo. HLO with a rounded configuration was counted as the surviving while the organoids with out of shape is counted as dead. To assess the

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survival rate of OA treated HLO at the same time point, 3D cell titer glo assay was used (Promega, Wi, USA).

Statistics and reproducibility. Statistical analysis was performed using unpaired two-tailed Student's t-test, Dunn-Holland-Wolfe test, or Welch's t-test. Results were shown as mean $\pm$ s.e.m.; P values < 0.05 were considered statistically significant. N-value refers to biologically independent replicates, unless noted otherwise.

All percentages and ratios are calculated by weight unless otherwise indicated.

All percentages and ratios are calculated based on the total composition unless otherwise indicated.

It should be understood that every maximum numerical limitation given throughout this specification includes every lower numerical limitation, as if such lower numerical limitations were expressly written herein. Every minimum numerical limitation given throughout this specification will include every higher numerical limitation, as if such higher numerical limitations were expressly written herein. Every numerical range given throughout this specification will include every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein.

The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as "20 mm" is intended to mean "about 20 mm."

Every document cited herein, including any cross referenced or related patent or application, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

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While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

What is claimed is:

1. A method of treating or reducing the occurrence of a steatohepatitis disorder, including NASH, parenteral nutrition associated liver disease (PNALD), and a genetic form of liver disease, comprising the step of administering a composition comprising obeticholic acid and total parenteral nutrition or peripheral parenteral nutrition to an individual in need thereof, wherein the total parenteral nutrition or peripheral parenteral nutrition comprises a lipid emulsion, and wherein said composition is administered parenterally.

2. The method of claim 1, wherein said composition is administered in an amount sufficient to reduce or prevent the occurrence of liver cell fibrosis.

3. The method of claim 1 wherein said composition is administered prior to, following, or during administration of parenteral nutrition to said individual in need thereof.

4. The method of claim 1 wherein said composition further comprises a buffer.

5. The method of claim 1 wherein said composition further comprises a carrier.

6. The method of claim 1 wherein said composition is administered intravenously.

7. A parenteral composition comprising

a. total parenteral nutrition (TPN) or peripheral parenteral nutrition (PPN), wherein the total parenteral nutrition or peripheral parenteral nutrition comprises a lipid emulsion; and

b. obeticholic acid.

8. A kit comprising a container, the composition of claim 7 in a dosage form disposed in the container, and instructions for administering said dosage form to an individual diagnosed or suspected of having or developing PNALD.

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